

Addition Problem Solutions

$$1. a) \frac{dy}{dx} = y + x \quad p(x) = -1$$

$$\frac{dy}{dx} - y = x \quad q(x) = x$$

$$\int p(x) dx = \int -1 dx = -x$$

$$I(x) = e^{\int p(x) dx} = e^{-x}$$

$$e^{-x} \frac{dy}{dx} - e^{-x} y = x e^{-x}$$

$$\frac{d}{dx}(e^{-x} y) = x e^{-x}$$

$$e^{-x} y = \int x e^{-x} dx$$

$$u = x \quad dv = e^{-x} dx$$

$$du = dx \quad v = -e^{-x}$$

$$= -x e^{-x} + \int e^{-x} dx$$

$$e^{-x} y = -x e^{-x} - e^{-x} + C$$

$$\boxed{y = -x - 1 + C e^{-x}}$$

$$b) x' = 2x + t e^{2t} \quad p(t) = -2$$

$$x' - 2x = t e^{2t} \quad q(t) = t e^{2t}$$

$$\int p(t) dt = \int -2 dt = -2t$$

$$I(t) = e^{\int p(t) dt} = e^{-2t}$$

$$e^{-2t} x' - 2x e^{-2t} = t$$

$$(e^{-2t} x)' = t$$

$$e^{-2t} x = \int t dt$$

$$e^{-2t} x = \frac{1}{2} t^2 + C$$

$$\boxed{x = \frac{1}{2} t^2 e^{2t} + C e^{2t}}$$

$$c) (1+t)y' - ty = t + t^2$$

$$p(t) = \frac{t}{1+t}$$

$$y' - \frac{t}{1+t}y = \frac{t+t^2}{1+t}$$

$$q(t) = \frac{t+t^2}{1+t}$$

$$\int p(t)dt = \int \frac{-t}{1+t} dt$$

$$u = 1+t \Rightarrow t = u-1$$

$$du = dt$$

$$= \int \frac{1-u}{u} du$$

$$= \int \frac{1}{u} - 1 du$$

$$= \ln|u| - u$$

$$= \ln|1+t| - 1 - t$$

$$I(t) = e^{\int p(t)dt} = e^{\ln|1+t| - 1 - t} = (1+t)e^{-(1+t)}$$

$$(1+t)e^{-(1+t)}y' - (1+t)e^{-(1+t)}\frac{t}{1+t}y = (1+t)e^{-(1+t)}\frac{t+t^2}{1+t}$$

$$((1+t)e^{-(1+t)}y)' = (t+t^2)e^{-(1+t)}$$

$$(1+t)e^{-(1+t)}y = \int (t+t^2)e^{-(1+t)} dt$$

	u	dv
(+)	$t+t^2$	$e^{-(1+t)}$
(-)	$1+2t$	$-e^{-(1+t)}$
(+)	2	$e^{-(1+t)}$
		$-e^{-(1+t)}$

$$= -(t+t^2)e^{-(1+t)} - (1+2t)e^{-(1+t)} - 2e^{-(1+t)} + C$$

$$= -(t+t^2+1+2t+2)e^{-(1+t)} + C$$

$$= -(t^2+3t+3)e^{-(1+t)} + C$$

$$y = \frac{-(t^2+3t+3)}{t+1} + \frac{Ce^{-t}}{t+1}$$

$$d) tx' + (1+t)x = e^{-t} \sin 2t$$

$$p(t) = \frac{1+t}{t}$$

$$x' + \frac{1+t}{t} x = \frac{1}{t} e^{-t} \sin 2t$$

$$q(t) = \frac{1}{t} e^{-t} \sin 2t$$

$$\int p(t) dt = \int \frac{1+t}{t} dt$$

$$= \int \frac{1}{t} + 1 dt$$

$$= \ln |t| + t$$

$$I(t) = e^{\ln |t| + t} = te^t$$

$$te^t x' + \frac{1+t}{t} te^t x = te^t \frac{1}{t} e^{-t} \sin 2t$$

$$(te^t x)' = \sin 2t$$

$$te^t x = \int \sin 2t dt$$

$$= -\frac{1}{2} \cos 2t + C$$

$$x = -\frac{1}{2t} e^{-t} \cos 2t + \frac{C}{t} e^{-t}$$

$$2.a) V(t) = 500 + 3t - 5t$$

$$V(t) = 500 - 2t$$

$$V(1) = 500 - 2(1)$$

$$V(10) = 500 - 2(10)$$

$$V(50) = 500 - 2(50)$$

$$V(1) = 498 \text{ gal.}$$

$$V(10) = 480 \text{ gal.}$$

$$V(50) = 400 \text{ gal.}$$

$$0 = 500 - 2t$$

$$t = 250 \text{ min.}$$

$$b) \frac{dS}{dt} = \left[\text{rate} \right]_{\text{in}} - \left[\text{rate} \right]_{\text{out}}$$

$$= (2 \text{ lbs./gal.})(3 \text{ gal./min.}) - \left[S(t)/(500-2t) \right](5 \text{ gal./min.})$$

$$\boxed{\frac{dS}{dt} = 6 - \frac{5}{500-2t} S}$$

$$\frac{dS}{dt} + \frac{5}{500-2t} S = 6 \quad p(t) = \frac{5}{500-2t} \quad q(t) = 6$$

$$\int p(t) dt = \int \frac{5}{500-2t} dt$$

$$= -\frac{5}{2} \ln |500-2t|$$

$$I(t) = e^{\int p(t) dt} = e^{\ln |500-2t|^{-5/2}}$$

$$= (500-2t)^{-5/2}$$

$$(500-2t)^{-5/2} \frac{dS}{dt} + (500-2t)^{-5/2} \frac{5}{500-2t} S = 6(500-2t)^{-5/2}$$

$$\frac{d}{dt} [(500-2t)^{-5/2} S] = 6(500-2t)^{-5/2}$$

$$(500-2t)^{-5/2} S = \int 6(500-2t)^{-5/2} dt$$

$$= 6 \cdot \frac{-2}{3} \cdot \frac{-1}{2} (500-2t)^{-3/2} + C$$

$$= 2(500-2t)^{-3/2} + C$$

$$S(t) = 2(500-2t) + C(500-2t)^{5/2}$$

$$S(0) = 2(500-2(0)) + C(500-2(0))^{5/2}$$

$$25 = 1000 + 500^{5/2} C$$

$$C = \frac{-975}{500^{5/2}}$$

$$\boxed{S(t) = 2(500-2t) - \frac{975}{500^{5/2}} (500-2t)^{5/2}}$$

$$c) S(10) = 2(500 - 2(10)) - \frac{975}{500^{5/2}} (500 - 2(10))^{5/2}$$

$$S(10) \approx 79.6 \text{ lbs.}$$

$$S(200) = 2(500 - 2(200)) - \frac{975}{500^{5/2}} (500 - 2(200))^{5/2}$$

$$S(200) \approx 182.6 \text{ lbs.}$$

$$S(100) = 2(500 - 2(100)) - \frac{975}{500^{5/2}} (500 - 2(100))^{5/2}$$

$$S(100) \approx 328.1 \text{ lbs.}$$

$$S(240) = 2(500 - 2(240)) - \frac{975}{500^{5/2}} (500 - 2(240))^{5/2}$$

$$S(240) \approx 39.7 \text{ lbs.}$$

$$d) S(t) = 2(500 - 2t) - \frac{975}{500^{5/2}} (500 - 2t)^{5/2}$$

$$S'(t) = -4 - \frac{975}{500^{5/2}} \cdot \frac{5}{2} (500 - 2t)^{3/2} \cdot (-2)$$

$$-4 + \frac{4875}{500^{5/2}} (500 - 2t)^{3/2} = 0$$

$$\frac{4875}{500^{5/2}} (500 - 2t)^{3/2} = 4$$

$$(500 - 2t)^{3/2} = \frac{4 \cdot 500^{5/2}}{4875}$$

$$500 - 2t = \left(\frac{4 \cdot 500^{5/2}}{4875} \right)^{2/3}$$

$$t = 250 - \frac{1}{2} \left(\frac{4 \cdot 500^{5/2}}{4875} \right)^{2/3}$$

$$t \approx 112 \text{ min}$$

$$S(112) \approx 331.3 \text{ lbs.}$$